Learning from General Language Feedback

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Abstract

A language teacher can comment on an agent's execution without supplying a corrected trajectory, an instruction to imitate, or a scalar reward. We formulate *General Language Feedback Training* (GLFT), which first behavior-clones observed executions conditioned on their original requests and hindsight feedback, then conditions this model on independently sampled *ideal feedback* and distills the resulting executions into a request-only policy. In a finite, deterministic sequential environment with exact population conditional imitation and sufficient initial support, the procedure implements a multiplicative posterior reweighting. Its induced feedback distribution converges to a calibrated ideal distribution at last-iterate rate $O(1/T)$ in KL divergence. Feedback matching alone does not imply task success. We derive a sufficient and quantitatively checkable *optimality-separating feedback statistic*: its mean is constant on optimal executions and uniformly smaller on suboptimal executions. This yields convergence of the deployed policy to the set of optimal behaviors, with $O(T^{-1/2})$ error. We construct such a channel from *truthful, randomly selected task-requirement reports*, permitting omitted clauses, paraphrases, arbitrary clause order, and multiple correct executions. The construction requires no language–action alignment oracle and no erroneous teacher labels; its feedback can nevertheless be ambiguous on any given episode. We further identify the exact distillation gradient with the gradient of feedback KL, and give a conditional two-timescale stochastic-approximation theorem for continuous auxiliary and clean-policy SGD. It guarantees posterior tracking and actor stationarity under strong auxiliary optimization and smoothness assumptions, not global neural-policy optimality. We delineate ideal-feedback access, realizability, coverage, and gradient-estimation costs; no polynomial-interaction or scalar-RL separation follows.

Keywords: language feedback; conditional imitation; policy distillation; population convergence; feedback identification; sequential decision making.

1 Introduction and scope

Suppose a user asks an assistant to send a particular report to Alice and give Bob view-only access. The assistant may send the wrong report, or grant edit permission. Afterward, an observer may say "The recipient was right, but the file version and permissions were wrong." That feedback is informative, but it is not a new executable request and it does not provide the correct action sequence. Direct hindsight-request relabeling would therefore be inappropriate.

General Language Feedback Training (GLFT) treats the feedback as *evidence about which executions occurred*. It learns the conditional distribution of its own executions given the original request and resulting feedback. It then queries that conditional model using language associated with fully satisfactory behavior and distills the generated executions into the ordinary policy, which uses only the request at deployment.

The population theory has two logically distinct targets. The algorithm first converges in *feedback space*: would an external teacher describe its executions as it would describe an optimal execution? To conclude *behavioral optimality*, the feedback must make it impossible for an inappropriate mixture

of executions to mimic the ideal feedback distribution. We develop a quantitative, constructive sufficient condition for this second step, rather than presupposing optimality identification.

Convergence is not a per-round improvement guarantee. GLFT does not include a reward-based policy-improvement operator. Even under the optimality-separation condition used for its convergence theorem, a single ideal-feedback-conditioned distillation update can reduce task success. The proof instead follows a decreasing *feedback-distribution* KL potential and a comparator KL potential; Section 9 gives an explicit example satisfying the theorem's assumptions.

Relation to existing approaches. Hindsight experience replay [2] and hindsight-instruction relabeling [3, 4] use achieved outcomes or inferred instructions to relabel training examples. GLFT instead keeps the original request fixed and conditions a separate execution model on *post-execution evaluative descriptions*; this distinction matters when a comment such as "permission was too broad" is not itself a valid replacement instruction. The exact population operator below is also a latent-variable likelihood/EM update [1]; we do not claim that its multiplicative form is new. The contribution of the analysis is its feedback-channel interpretation, explicit identification conditions, and relation to continuously trained policies.

Scope. This document supplies the formulation, exact-population results, an operational feedback construction, and proofs. It is deliberately separate from the earlier proposed finite-interaction GLFT-versus-RLOO separation; that claim requires its own grounding and oracle-access audit. An additional gradient-based analysis states explicitly which guarantees survive incomplete, incremental optimization; all original exact-population assumptions and results are retained.

2 Sequential problem and feedback

2.1 Requests, executions, and optimality

Let $D \sim q^\star$ be a request. For a fixed D, an agent interacts with a finite-horizon, deterministic controlled environment with fixed initial state, horizon H, and finite action and observation sets. An execution $X = (o_1, a_1, \ldots, o_H, a_H)$ includes the complete observable history. Write $\mathcal{X}(D)$ for feasible executions. The deployed request-conditioned, potentially history-dependent policy $\pi_t(a_h \mid \mathcal{H}_h, D)$ induces the *complete-execution distribution* $p_t(x \mid D)$ on $\mathcal{X}(D)$, where $\mathcal{H}_h$ is the observed history before decision h. Thus p_t is not a separately trained model: $X \sim p_t(\cdot \mid D)$ means rolling out π_t in the environment under request D. With deterministic dynamics and a fixed initial state, $p_t(x \mid D) = \prod_{h=1}^{H} \pi_t(a_h \mid \mathcal{H}_h, D)$ for any feasible execution x.

Each request has a nonempty set $\mathcal{C}(D) \subseteq \mathcal{X}(D)$ of optimal executions. Multiple correct execution orders, tool-use sequences, and equivalent plans may be optimal. We seek

$$p_t(\mathcal{C}(D) \mid D) \longrightarrow 1, \tag{1}$$

not convergence to a user-specified distribution over correct trajectories. For finite spaces, this is equivalent to approaching the set $\mathcal{P}^\star(D) = \{p \in \Delta(\mathcal{X}(D)) : p(\mathcal{C}(D)) = 1\}$ in total variation. For a binary success objective, it also means optimal expected reward.

Why deterministic dynamics appear here. We require the posterior distribution over whole trajectories, conditional on feedback, to be realizable by a policy acting in the original environment. With independent uncontrollable transition noise, conditioning on hindsight success generally alters the law of exogenous events and is not executable by the agent. The exact posterior-reproduction claim therefore does not hold in arbitrary stochastic MDPs without additional structure.

2.2 Feedback channel and ideal-feedback generator

After execution x, a teacher draws free-form hindsight feedback $f \in \mathcal{F}(D)$ using a channel

$$f \sim F_D(\cdot \mid x). \tag{2}$$

We use a finite feedback alphabet for the proofs. An element f can be a sentence or a sequence of raw text tokens; the analysis does not provide a teacher parser or a correspondence between its clauses and actions. The teacher need not give action labels, rewards, corrections, or demonstrations. Its descriptions may be stochastic, abbreviated, reordered, or paraphrased.

For any execution distribution p, define its feedback marginal

$$m_p(f \mid D) := \sum_{x \in \mathcal{X}(D)} p(x \mid D) F_D(f \mid x). \tag{3}$$

We write $m_t = m_{p_t}$ for a fixed request. The second training-time resource is a sampler

$$f^\star \sim G^\star_D, \tag{4}$$

which can be used without observing an optimal execution. It is an assumption about supervision available for distillation, not something the algorithm has proved how to learn. It may be stochastic and need not output one canonical "success" sentence.

Definition 2.1 (Calibrated ideal feedback)**.** *The ideal-feedback generator is calibrated if, for every supported request D, there exists an optimal witness distribution $\bar{p}_D \in \mathcal{P}^\star(D)$ such that*

$$G^\star_D(f) = \sum_x \bar{p}_D(x) F_D(f \mid x) = m_{\bar{p}_D}(f \mid D). \tag{5}$$

The witness need not be observed, unique, or reproduced by the learned policy. Importantly, $G^\star_D$ does not itself reveal a corrected action sequence. Nevertheless, access to it is additional training-time information relative to receiving only on-policy hindsight feedback.

Definition 2.2 (Coverage and exact population realizability)**.** *For every relevant D, assume (i)* $\mathrm{supp}(\bar{p}_D) \subseteq \mathrm{supp}(p_0(\cdot \mid D))$*; (ii) each feedback message with $G^\star_D(f) > 0$ is covered by the induced training joint; and (iii) the feedback-conditioned and clean policy classes, trained by exact population conditional maximum likelihood, can realize the corresponding full-trajectory conditional distributions in the unchanged deterministic environment.*

Part (ii) follows from (i) and calibration at initialization, and is propagated by the exact population operator below; it is stated to highlight the operational requirement. Finite-batch MLE and restricted linear or neural policy classes are not covered by this assumption.

3 The GLFT procedure

The exact population procedure has two equivalent descriptions: conditional imitation followed by distillation, and the induced request-wise distributional operator in Algorithm 1. The algorithm displays the *operator analyzed by the convergence theorems*; it is not finite-sample training pseudocode. We fix a supported request D in the algorithm because the exact update and its convergence hold separately for each request. For shared policies trained with population expectations over $D \sim q^\star$, exact conditional maximum likelihood identifies the required request-wise conditionals $q^\star$-almost

everywhere under shared-policy realizability. It need not identify them on $q^\star$-null requests. When $q^\star$ is discrete, this conclusion holds at each request assigned positive probability; the fixed-request analysis below applies to any D satisfying its assumptions.

Stage I: feedback-conditioned behavior cloning. The on-policy population joint law of requests, executions, and hindsight feedback is defined by

$$D \sim q^\star, \qquad X \sim p_t(\cdot \mid D), \qquad F \sim F_D(\cdot \mid X). \tag{6}$$

Fit an auxiliary policy $\pi_t^{\mathrm{fb}}(a_h \mid \mathcal{H}_h, D, F)$ to the executed actions using conditional maximum likelihood,

$$\max_{\pi^{\mathrm{fb}}} \ \mathbb{E}_{D,X,F} \sum_{h=1}^{H} \log \pi^{\mathrm{fb}}(a_h \mid \mathcal{H}_h, D, F). \tag{7}$$

The original request is not replaced by feedback. In particular, "the permission was too broad" need not itself be an executable instruction.

Stage II: ideal-feedback sampling and distillation. The ideal-feedback-conditioned population law is induced by $D \sim q^\star$, $F^\star \sim G_D^\star$, and the Stage-I conditional execution distribution $X^+ \sim \pi_t^{\mathrm{fb}}(\cdot \mid D, F^\star)$. Exact distillation maximizes

$$\max_{\pi} \ \mathbb{E}_{D,F^\star,X^+} \sum_{h=1}^{H} \log \pi(a_h \mid \mathcal{H}_h, D). \tag{8}$$

Only $\pi_{t+1}(a_h \mid \mathcal{H}_h, D)$ acts at the next interaction round and at deployment. It receives no ideal or hindsight feedback at test time.

Algorithm 1 Population GLFT: exact request-wise distributional update

Input: Fixed supported request D; initial execution law $p_0(\cdot \mid D)$ induced by π_0; feedback channel F_D (population object); ideal-feedback law $G_D^\star$; rounds T

1: **for** $t = 0, \dots, T-1$ **do**

2: **Feedback marginal:** For every $f \in \mathcal{F}(D)$, compute
$m_t(f \mid D) \leftarrow \sum_{x \in \mathcal{X}(D)} p_t(x \mid D) F_D(f \mid x)$.

3: **Conditional execution law:** For each f with $G_D^\star(f) > 0$, set
$r_t(x \mid D, f) \leftarrow \dfrac{p_t(x \mid D) F_D(f \mid x)}{m_t(f \mid D)} \quad (x \in \mathcal{X}(D))$.

4: **Ideal-feedback update:** For every $x \in \mathcal{X}(D)$, set
$p_{t+1}(x \mid D) \leftarrow \sum_{f : G_D^\star(f) > 0} G_D^\star(f)\, r_t(x \mid D, f)$.

5: **end for**

Output: $p_T(\cdot \mid D)$, induced by a realizable request-only policy π_T under exact population distillation

Algorithm 1 is a closed-form description of the exact population operator, not a requirement that the learner know or evaluate the feedback kernel F_D, nor a recipe involving a single training triple or a finite number of gradient updates. Its loops over x and f specify a mathematical mapping rather than operations counted toward a finite-interaction complexity bound. The learner's Stage-I access to F_D is through hindsight feedback generated after executions; Stage-II uses samples from $G_D^\star$. Calibration and initial coverage guarantee $m_t(f \mid D) > 0$ wherever $G_D^\star(f) > 0$. Under Definition 2.2, its conditional execution law r_t is exactly the Stage-I population-MLE optimum of (7); its updated execution law p_{t+1} is exactly the Stage-II population-MLE optimum of (8). Thus the request-only

π_T in the output is the policy realizing p_T, not an extra model maintained alongside it. No corrected trajectory, scalar reward, or language–action alignment label is assumed. Online GLFT, introduced later, instead gives the stochastic-gradient procedure.

4 Exact population dynamics

As in Algorithm 1, fix a supported request D and suppress it in the notation. The equivalent conditional-imitation and distillation population objectives in (7)–(8) may average over $q^\star$ when fitting shared policies. Under exact conditional realizability, a shared-policy population MLE induces this distributional update for $q^\star$-almost every D; for discrete $q^\star$, this includes each positive-mass request. The fixed-D operator and the following theorems also hold pointwise for any request satisfying their assumptions, irrespective of the joint-training interpretation, and their bounds may be averaged under $q^\star$.

Lemma 4.1 (Posterior and multiplicative update)**.** *Under Definition 2.2, the population optimum of* (7) *realizes*

$$r_t(x \mid f) = \frac{p_t(x)F(f \mid x)}{m_t(f)} \qquad (m_t(f) > 0). \tag{9}$$

Exact distillation (8) *yields*

$$\begin{aligned} p_{t+1}(x) &= \sum_{f:G^\star(f)>0} G^\star(f) r_t(x \mid f) = p_t(x) W_t(x), \\ W_t(x) &:= \sum_{f:G^\star(f)>0} F(f \mid x) \frac{G^\star(f)}{m_t(f)}. \end{aligned} \tag{10}$$

The right-hand side is normalized and preserves support of every calibrated optimal witness $\bar{p}$.

Proof. The Stage-I joint law is $p_t(x)F(f \mid x)$, hence its exact conditional MLE is the Bayes posterior in (9). Because deterministic transition dynamics allow arbitrary feasible history-dependent action distributions to be implemented, the exact conditional policy realizes this posterior. Stage-II conditional sampling and exact cloning marginalize over $G^\star$, giving (10). Summing over x gives $\sum_x p_t(x)W_t(x) = \sum_f G^\star(f) = 1$. If $\bar{p}(x) > 0$, then every f with $F(f \mid x) > 0$ has $G^\star(f) > 0$ by $G^\star = F\bar{p}$. Such an f also has $m_t(f) > 0$ whenever $p_t(x) > 0$. Consequently $W_t(x) > 0$, and the witness support is preserved inductively. □

Theorem 4.2 (Population feedback convergence)**.** *Assume calibrated ideal feedback, coverage, and exact population conditional realizability. Let*

$$\Phi_t := D_{\mathrm{KL}}(G^\star \| m_t), \qquad V_t := D_{\mathrm{KL}}(\bar{p} \| p_t). \tag{11}$$

Then, for all $t \geq 0$,

$$\Phi_t - \Phi_{t+1} \geq D_{\mathrm{KL}}(p_{t+1} \| p_t) \geq 0, \qquad V_t - V_{t+1} \geq \Phi_t. \tag{12}$$

Consequently Φ_t is nonincreasing, and

$$D_{\mathrm{KL}}(G^\star \| m_T) \leq \frac{D_{\mathrm{KL}}(\bar{p} \| p_0)}{T+1} \qquad (T \geq 0). \tag{13}$$

Proof. The first inequality is the EM-type monotonicity argument. Writing $r_t(x \mid f)$ for the Bayes posterior, observe that

$$\frac{m_{t+1}(f)}{m_t(f)} = \sum_x r_t(x \mid f) W_t(x).$$

Jensen's inequality and then a change of order give

$$\begin{aligned}
\Phi_t - \Phi_{t+1} &= \sum_f G^\star(f) \log \frac{m_{t+1}(f)}{m_t(f)} \\
&\geq \sum_{f,x} G^\star(f) r_t(x \mid f) \log W_t(x) \\
&= \sum_x p_{t+1}(x) \log \frac{p_{t+1}(x)}{p_t(x)}.
\end{aligned}$$

For the second inequality, (10) and Jensen give

$$\begin{aligned}
V_t - V_{t+1} &= \sum_x \bar{p}(x) \log W_t(x) \\
&\geq \sum_{x,f} \bar{p}(x) F(f \mid x) \log \frac{G^\star(f)}{m_t(f)} \\
&= \sum_f G^\star(f) \log \frac{G^\star(f)}{m_t(f)} = \Phi_t.
\end{aligned}$$

All logarithms here are evaluated on the positive support of the corresponding distribution. Summing the second inequality over $t = 0, \ldots, T$ gives $\sum_{t=0}^{T} \Phi_t \leq V_0$; by monotonicity $(T+1)\Phi_T \leq \sum_{t=0}^{T} \Phi_t$. This proves (13). □

EM interpretation and scope of novelty. Treat x as a latent mixture component, $F(f \mid x)$ as its emission likelihood, and $G^\star(f)$ as the target distribution of observed feedback. The posterior $r_t(x \mid f)$ is the EM E-step, while $p_{t+1}(x) = \sum_f G^\star(f) r_t(x \mid f)$ is the exact mixture-weight M-step. Therefore the monotonicity mechanism is a standard EM-type likelihood argument [1]; the substantive behavioral guarantee additionally requires calibration and optimality-separating feedback.

What converges? We have proved convergence of the feedback marginal, not of the parameters of a particular neural policy or toward a preselected trajectory mixture. By Pinsker,

$$\mathrm{TV}(G^\star, m_T) \leq \sqrt{\frac{V_0}{2(T+1)}}. \tag{14}$$

Matching feedback does not, by itself, force correct behavior. *Nor does monotone decrease of Φ_t imply monotone increase of $p_t(\mathcal{C})$; the latter can temporarily decrease even when the sufficient conditions for eventual optimality hold.*

5 When does feedback matching imply optimality?

5.1 A sufficient statistical separation condition

Let $\mathcal{C} \subset \mathcal{X}$ be the optimal executions for the fixed request. The weakest identification property relevant to Theorem 4.2 is

$$m_p = G^\star \implies p(\mathcal{C}) = 1 \qquad \text{for every execution distribution } p \text{ on accessible support.} \tag{15}$$

It is insufficient to require merely $F(\cdot \mid x) \neq G^\star$ for each individual bad x: mixtures of bad executions may still reproduce $G^\star$.

Definition 5.1 (Optimality-separating feedback statistic)**.** *A feedback channel is γ-optimality separating for the request if there exists a statistic $s : \mathcal{F} \to [0,1]$, a constant $a \in [0,1]$, and $\gamma > 0$ such that*

$$\mathbb{E}[s(F) \mid X = x] = a \quad \forall x \in \mathcal{C}, \qquad \mathbb{E}[s(F) \mid X = x] \leq a - \gamma \quad \forall x \notin \mathcal{C}. \tag{16}$$

The statistic is a certificate used by the analysis: the GLFT algorithm is neither given s nor trained to predict it.

The shared optimal mean in (16) is important. Without it, mixtures of high-scoring optimal and low-scoring incorrect executions could cancel a feedback gap. The condition allows overlap between the language distributions of individual correct and incorrect executions.

Lemma 5.2 (Separation of every suboptimal mixture)**.** *If $G^\star$ is calibrated and* (16) *holds, then every execution distribution p satisfies*

$$\gamma p(\mathcal{C}^c) \leq \mathbb{E}_{G^\star} s - \mathbb{E}_{m_p} s \leq \mathrm{TV}(G^\star, m_p). \tag{17}$$

In particular (15) *holds.*

Proof. Calibration gives $\mathbb{E}_{G^\star} s = \mathbb{E}_{\bar{p}} \mathbb{E}[s \mid X] = a$. Averaging (16) under p gives $\mathbb{E}_{m_p} s \leq a - \gamma p(\mathcal{C}^c)$. For a statistic valued in $[0,1]$, its expectation difference under two laws is bounded by their total variation distance. □

Theorem 5.3 (Convergence to optimal behavior)**.** *Under the assumptions of Theorem 4.2 and the γ-optimality separating condition* (16)*, for every $T \geq 0$,*

$$p_T(\mathcal{C}^c) \leq \min\left\{1, \frac{1}{\gamma}\sqrt{\frac{D_{\mathrm{KL}}(\bar{p} \| p_0)}{2(T+1)}}\right\}. \tag{18}$$

Hence $p_T(\mathcal{C}) \to 1$. For binary reward $R(x) = \mathbf{1}\{x \in \mathcal{C}\}$, the expected reward tends to its optimal value 1*. The theorem does not imply convergence to any particular optimal mixture $p^\star$.*

Proof. Combine (17) with (14). □

5.2 What if the optimality gap becomes arbitrarily small?

Uniform $\gamma > 0$ can fail in continuous-quality tasks. Let $R : \mathcal{X} \to [0,1]$, $R^\star = \max_x R(x)$, and fix $\varepsilon > 0$. A weaker requirement is that there exist $s \in [0,1]$, a, and $\gamma_\varepsilon > 0$ such that $\mathbb{E}[s \mid x] \leq a$ for every execution, with equality on every optimal execution, and $\mathbb{E}[s \mid x] \leq a - \gamma_\varepsilon$ whenever $R(x) \leq R^\star - \varepsilon$. Repeating the argument gives

$$\Pr_{p_T}\{R(X) \leq R^\star - \varepsilon\} \leq \frac{1}{\gamma_\varepsilon}\sqrt{\frac{V_0}{2(T+1)}}. \tag{19}$$

This proves asymptotic concentration on ε-optimal outcomes for each fixed ε; it does not require a uniform gap over trajectories with arbitrarily small reward suboptimality.

6 A concrete feedback channel: truthful partial requirement reports

The optimality-separating condition need not be asserted abstractly. We now construct an interpretable feedback process that satisfies it, without requiring the teacher to diagnose every requirement on every execution.

6.1 Task-relative requirements and a partial auditor

For each request D, define $m = m(D) \geq 1$ verifiable requirements, with indicators

$$r_j(D, x) \in \{0, 1\}, \quad j = 1, \ldots, m, \qquad \mathcal{C}(D) = \{x : r_j(D, x) = 1 \ \forall j\}. \tag{20}$$

For an email-and-permissions task, requirements might include the recipient, attachment version, and permission level. Different trajectories may satisfy all requirements through different tool sequences.

After execution, the teacher selects an audit subset $S \subseteq \{1, \ldots, m\}$ from a distribution Q_D that depends on the request, *not* on the execution or its achieved truth values. Specifically,

$$\Pr(S = s \mid D, x) = Q_D(s), \qquad \kappa_j(D) := \Pr_{S \sim Q_D}(j \in S) \geq \kappa_{\min}(D) > 0. \tag{21}$$

The full-law independence in (21) is needed to calibrate a request-only ideal-feedback sampler for *every* optimal execution. Merely requiring execution-independent marginal inclusion rates $\kappa_j(D)$ would suffice for the score-gap calculation below, but not for that calibration claim. There is no requirement that inclusion events for different j be independent. The teacher then truthfully reports the satisfaction or violation of every $j \in S$, in arbitrary order and arbitrary semantically faithful language. Conditional on D, S and these truth values, the distribution of report wording and ordering does not depend on any other property of the execution. It may omit the other requirements entirely. Feedback chunks may be paraphrased, grouped, or shuffled; no action–chunk alignment is provided.

Example. Request: "Send the Q3 summary to Alice and give Bob read access." An execution sends Q3 to Alice but gives Bob edit rights. Depending on S, feedback might say "The document and recipient were correct," or "Bob received edit permission rather than read-only access." Both are accurate partial reports of the same execution.

An ideal-feedback sampler can be implemented at the level of the request: sample S by the same audit-selection procedure and verbalize the selected requirements as satisfied. When every requirement is achievable, this samples the feedback distribution of any optimal execution, without needing to construct that execution. Sampling ideal feedback still assumes access to accurate descriptions of satisfied requirements; it is not free information.

Proposition 6.1 (Truthful partial audits separate optimal behavior)**.** *Under* (20)–(21)*, suppose the language explicitly and correctly conveys the polarity of each reported requirement (though not its action alignment). Define*

$$s_D(F) := \frac{\#\{j \in S : \textit{requirement } j \textit{ is reported satisfied}\}}{m}. \tag{22}$$

Then $0 \leq s_D \leq 1$ *and*

$$\mathbb{E}[s_D(F) \mid D, X = x] = \frac{1}{m} \sum_{j=1}^{m} \kappa_j(D) r_j(D, x). \tag{23}$$

Consequently the channel satisfies Definition 5.1 with

$$a_D = \frac{1}{m}\sum_{j=1}^{m}\kappa_j(D), \qquad \gamma_D = \frac{\kappa_{\min}(D)}{m(D)}. \tag{24}$$

The ideal-feedback generator described above is calibrated for every optimal witness distribution.

Proof. By truthful reporting, the numerator in (22) is $\sum_j \mathbf{1}\{j \in S\} r_j(D, x)$. Taking expectations yields (23); independence across requirements is unnecessary. Every optimal execution has all $r_j = 1$ and therefore expectation a_D. Every suboptimal execution fails at least one requirement, causing a loss of at least $\kappa_{\min}/m$. For any optimal execution, every sampled requirement is described as satisfied. Since the *entire* audit-subset law is Q_D for every execution, and the wording kernel depends only on D, S, and the selected truth values, ideal-feedback sampling from the request reproduces exactly the teacher's conditional law for every optimal execution. This last conclusion would not follow from the marginal rates κ_j alone. □

Corollary 6.2 (Explicit optimality rate under truthful partial audits)**.** *Under the exact population assumptions and the partial-audit channel,*

$$p_T(\mathcal{C}(D)^c \mid D) \le \frac{m(D)}{\kappa_{\min}(D)}\sqrt{\frac{D_{\mathrm{KL}}(\bar{p}_D \| p_0(\cdot \mid D))}{2(T+1)}}. \tag{25}$$

The bound is understood as capped at one.

A less pessimistic graded-quality rate. Suppose all requirements have the same inclusion probability $\kappa > 0$ and quality is measured by the fraction satisfied,

$$R_D(x) := \frac{1}{m}\sum_{j} r_j(D, x), \qquad R_D^\star = 1. \tag{26}$$

Then (23) gives $\mathbb{E}[s_D \mid x] = \kappa R_D(x)$ and $\mathbb{E}_{G_D^\star} s_D = \kappa$. Therefore,

$$1 - \mathbb{E}_{p_T} R_D(X) \le \frac{1}{\kappa}\sqrt{\frac{D_{\mathrm{KL}}(\bar{p}_D \| p_0(\cdot \mid D))}{2(T+1)}}. \tag{27}$$

This quality bound has no explicit factor of m; exact all-requirements success is a stricter target than high average requirement satisfaction.

6.2 Why incomplete truthful feedback is sufficient

The teacher may omit the one failed requirement in a particular episode. Thus a suboptimal execution can receive feedback that is indistinguishable from a piece of ideal feedback, despite zero teacher misreporting. The proof does not require identifying optimality from a single message. It requires that any persistently violated requirement has a positive probability of being reported, producing a detectable difference in the feedback distribution. All calculations concern complete raw-language messages; they do not reveal segment–action alignment to the learner. The existence of s_D is a property of a semantically faithful reporting mechanism, not a supplied semantic parser.

7 A gradient view of GLFT

The exact-population theory above minimizes the mismatch between the teacher's ideal-feedback distribution and the feedback distribution induced by the clean policy. This objective also admits a direct gradient formulation, which motivates *incremental* clean-policy distillation rather than full inner-loop optimization. This section retains the fixed channel F_D, ideal-feedback law $G_D^\star$, full request D, and feasible deterministic executions of Sections 2–4. Parameter-space regularity conditions introduced below are *additional* to the original population theory.

Let π_θ be a differentiable clean policy with execution law $p_\theta(x \mid D)$ and feedback marginal $m_\theta(f \mid D)$. Let

$$J(\theta) := -\mathbb{E}_{D\sim q^\star} D_{\mathrm{KL}}(G_D^\star \| m_\theta(\cdot \mid D)), \qquad s_\theta(x, D) := \nabla_\theta \log p_\theta(x \mid D). \tag{28}$$

For the following identities assume that $G_D^\star(f) > 0$ implies $m_\theta(f \mid D) > 0$, that differentiation and expectation can be interchanged, and that the execution support and parameter derivatives are regular enough for $\nabla p_\theta(x \mid D) = p_\theta(x \mid D) s_\theta(x, D)$ on the relevant support. For policies with zero-probability executions, the score identities are evaluated only on their positive support; in the online algorithm, generated auxiliary trajectories must also admit a well-defined clean-policy score (e.g., a full-support clean softmax policy). In the deterministic environment,

$$s_\theta(x, D) = \sum_{h=1}^{H} \nabla_\theta \log \pi_\theta(a_h \mid \mathcal{H}_h, D).$$

Write $r_\theta(x \mid D, f)$ for the exact conditional execution posterior in Lemma 4.1, now using p_θ.

Proposition 7.1 (Policy-gradient and first-distillation identities)**.** *For every parameter value satisfying the preceding support and differentiability conditions,*

$$\nabla_\theta J(\theta) = \mathbb{E}_{\substack{D\sim q^\star,\, f^\star \sim G_D^\star \\ x^+ \sim r_\theta(\cdot \mid D, f^\star)}} s_\theta(x^+, D), \tag{29}$$

$$= \mathbb{E}_{\substack{D\sim q^\star,\, x\sim p_\theta(\cdot \mid D) \\ f\sim F_D(\cdot \mid x)}} \left[\frac{G_D^\star(f)}{m_\theta(f \mid D)} s_\theta(x, D) \right]. \tag{30}$$

Fix θ_t, and let $p_t^+(x \mid D) := \sum_f G_D^\star(f) r_{\theta_t}(x \mid D, f)$. Holding this pseudo-trajectory distribution fixed, define the conditional-distillation surrogate

$$L_t(\theta) := \mathbb{E}_{D\sim q^\star,\, x^+ \sim p_t^+(\cdot \mid D)} \log p_\theta(x^+ \mid D).$$

Then $\nabla L_t(\theta_t) = \nabla J(\theta_t)$. The identity does not *assert equality of these two gradients at arbitrary $\theta \neq \theta_t$.*

Proof. For each request, differentiating the forward KL objective gives

$$\begin{aligned}
\nabla J_D(\theta) &= \sum_f \frac{G_D^\star(f)}{m_\theta(f \mid D)} \nabla m_\theta(f \mid D) \\
&= \sum_{f,x} \frac{G_D^\star(f)}{m_\theta(f \mid D)} F_D(f \mid x) p_\theta(x \mid D) s_\theta(x, D) \\
&= \sum_f G_D^\star(f) \sum_x r_\theta(x \mid D, f) s_\theta(x, D).
\end{aligned}$$

The first and second expressions give (29) and (30). Since p_t^+ is treated as fixed when differentiating L_t, its score expectation evaluated at θ_t is exactly (29). □

Interpretation. At the current actor, *one* supervised distillation gradient with fresh, exact ideal-feedback-conditioned rollouts is an unbiased stochastic gradient of a fixed mathematical objective J. This is an equality of *first derivatives* at the current parameter value; it is not an equality of the full supervised objective and J. Once the actor changes, an unchanged distillation batch becomes stale; repeated steps on the same frozen surrogate need not follow ∇J.

Proposition 7.2 (Gradient bias from an approximate feedback-conditioned model)**.** *Let $r_\phi(\cdot \mid D, f)$ be the trajectory distribution induced by a learned feedback-conditioned policy. If $\|s_\theta(x, D)\| \le B$, define*

$$\delta(\theta, \phi) := \mathbb{E}_{D\sim q^\star,\, f\sim G^\star_D} \mathrm{TV}\left(r_\phi(\cdot \mid D, f), r_\theta(\cdot \mid D, f)\right).$$

The mean clean-policy score from auxiliary ideal-feedback rollouts, denoted $g(\theta, \phi)$, satisfies

$$\|g(\theta, \phi) - \nabla J(\theta)\| \le 2B\, \delta(\theta, \phi). \tag{31}$$

Proof. For any bounded vector-valued function u with $\|u\| \le B$, the difference in its expectations under two probability laws has norm at most $2B\,\mathrm{TV}(P, Q)$. Apply this with $u = s_\theta$, then average over D, f. □

7.1 Exact natural gradient on the trajectory simplex

There is also a geometric interpretation of the original GLFT update. For a fixed request and an execution distribution p in the relative interior of its supported face, define

$$W_p(x) := \sum_f F(f \mid x)\frac{G^\star(f)}{m_p(f)}, \qquad J(p) := -D_{\mathrm{KL}}(G^\star \| m_p).$$

Then $\partial J/\partial p(x) = W_p(x)$ and $\sum_x p(x)W_p(x) = 1$. The Fisher–Rao natural-gradient tangent vector on the trajectory simplex is

$$(\widetilde{\nabla} J(p))(x) = p(x)\big(W_p(x) - 1\big). \tag{32}$$

Thus the exact GLFT operator from Lemma 4.1 is precisely the *unit explicit-Euler step* $\mathcal{T}(p) = p + \widetilde{\nabla} J(p)$. This is not an identity between neural-parameter SGD and Fisher–Rao natural gradient.

Proposition 7.3 (Convergence of exact damped trajectory-space GLFT)**.** *Under the original exact-population assumptions of Theorem 4.2, let $\bar{p}$ be a calibrated, initially covered optimal witness. Fix $0 < \eta \le 1$ and define*

$$p_{t+1} = (1 - \eta)p_t + \eta\mathcal{T}(p_t), \quad \Phi_t = D_{\mathrm{KL}}(G^\star \| m_{p_t}), \quad V_t = D_{\mathrm{KL}}(\bar{p} \| p_t).$$

Then

$$\Phi_{t+1} \le \Phi_t, \quad V_t - V_{t+1} \ge \eta\Phi_t, \quad \Phi_T \le \frac{V_0}{\eta(T + 1)}. \tag{33}$$

If the original γ-optimality-separating condition holds, then

$$p_T(\mathcal{C}^c) \le \frac{1}{\gamma}\sqrt{\frac{V_0}{2\eta(T + 1)}}. \tag{34}$$

Proof. Both $p \mapsto D_{\mathrm{KL}}(G^\star \| m_p)$ and $p \mapsto D_{\mathrm{KL}}(\bar{p} \| p)$ are convex. Hence each potential at p_{t+1} is at most $(1 - \eta)$ times its value at p_t plus η times its value at $\mathcal{T}(p_t)$. Theorem 4.2 gives $\Phi(\mathcal{T}(p_t)) \le \Phi_t$ and $V_t - V(\mathcal{T}(p_t)) \ge \Phi_t$. Summing the second inequality and using the first to make Φ_t nonincreasing gives (33). The original feedback-statistic bound and Pinsker yield (34). □

Scope. The damped update operates on the *full trajectory simplex*; it may require exponentially large representations. It does not show that an ordinary neural step realizes this update, nor that its gradients can be estimated efficiently.

8 Online GLFT: incremental SGD for both policies

We next replace full inner-loop optimization of the two supervised stages by continuous stochastic gradients, while preserving the original policy roles and access to feedback. The resulting convergence theorem is *conditional*: the auxiliary loss is required to have optimization geometry strong enough to track its moving posterior, and the clean policy is shown to reach stationary points, not necessarily globally correct behavior.

8.1 Algorithm and supervised auxiliary objective

For differentiable auxiliary parameters ϕ, let $r_\phi(x \mid D, f)$ be the *executed trajectory law* of $\pi_\phi(a \mid \mathcal{H}, D, f)$ in the unchanged deterministic environment. Its population conditional behavioral-cloning loss is

$$\ell(\phi;\theta) := \mathbb{E}_{\substack{D\sim q^\star,\, X\sim p_\theta(\cdot|D) \\ F\sim F_D(\cdot|X)}}[-\log r_\phi(X \mid D, F)]. \tag{35}$$

The loss is just negative Stage-I conditional likelihood from Equation (7), expressed at the trajectory level. No feedback parser, scalar reward, or action-alignment oracle is introduced.

Algorithm 2 Online GLFT: ordinary SGD for both policies

Input: $q^\star$, F_D, $G^\star_D$; clean parameters θ_0; auxiliary parameters ϕ_0; step sizes $\{\alpha_t, \beta_t\}$

1: **for** $t = 0, 1, 2, \ldots$ **do**
2: Sample $D_t \sim q^\star$, roll out π_{θ_t} to obtain X_t, and obtain hindsight feedback $F_t \sim F_{D_t}(\cdot \mid X_t)$.
3: $\widehat{\ell}^{\mathrm{fb}}_t(\phi) \leftarrow -\sum_{h=1}^{H} \log \pi^{\mathrm{fb}}_\phi(a^{(t)}_h \mid \mathcal{H}^{(t)}_h, D_t, F_t)$.
4: **SGD:** $\phi_{t+1} \leftarrow \phi_t - \beta_t \nabla_\phi \widehat{\ell}^{\mathrm{fb}}_t(\phi_t)$.
5: Independently sample $D'_t \sim q^\star$ and ideal feedback $F^\star_t \sim G^\star_{D'_t}$.
6: Roll out $\pi^{\mathrm{fb}}_{\phi_{t+1}}(\cdot \mid \mathcal{H}_h, D'_t, F^\star_t)$ to obtain X^+_t.
7: $\widehat{\ell}^{\mathrm{clean}}_t(\theta) \leftarrow -\sum_{h=1}^{H} \log \pi_\theta(a^{+,(t)}_h \mid \mathcal{H}^{+,(t)}_h, D'_t)$.
8: **SGD:** $\theta_{t+1} \leftarrow \theta_t - \alpha_t \nabla_\theta \widehat{\ell}^{\mathrm{clean}}_t(\theta_t)$ ▷ hold X^+_t fixed
9: **end for**

Output: Continuously deployable clean policy π_{θ_t}

Algorithm 2 uses *ordinary stochastic gradient descent* on two negative log-likelihood losses; each iteration uses one freshly sampled execution–feedback triple and one freshly sampled ideal-feedback-conditioned execution (or independent minibatches). The first update trains the auxiliary model to imitate actions conditional on hindsight feedback; the second trains the request-only policy to imitate the generated execution. The generated execution X^+_t is held fixed when differentiating the clean-policy loss. The trajectory law $p_{\theta_t}(\cdot \mid D)$ denotes rollouts of π_{θ_t}, not a separately maintained density model. Auxiliary data must reflect the *current* clean-policy joint: naive mixing of old, unweighted replay data changes the population loss.

Projection in the theorem, not the practical pseudocode. To control the auxiliary iterates, Theorem 8.1 studies the *projected* variant of the auxiliary SGD step, $\phi_{t+1} = \Pi_K[\phi_t - \beta_t \nabla_\phi \widehat{\ell}^{\mathrm{fb}}_t(\phi_t)]$, for a compact convex K. Theorem 8.1 applies to Algorithm 2 as written only when the unprojected auxiliary iterates remain in K (so projection has no effect) and all the theorem's other assumptions hold. It does not automatically guarantee convergence for unconstrained neural-network SGD.

8.2 A conditional two-timescale convergence theorem

The following assumptions go beyond the original exact-population GLFT theorem. They are sufficient for the projected auxiliary-SGD variant of Algorithm 2; for unprojected SGD they suffice only if the projection would remain inactive along its trajectory. These are not claims about all neural language models or about Section 6's partial-audit construction.

(A1) Smooth bounded actor and compatible support. The unprojected actor iterates stay almost surely in a compact region Θ_0, and J has an L_J-Lipschitz gradient on a convex neighborhood containing every update segment. For each parameter pair visited, all trajectories that can be sampled from the ideal-feedback-conditioned auxiliary policy have positive clean-policy probability, so their clean-policy scores are well defined and obey $\|s_\theta(x, D)\| \le B$. The same score bound holds on the current clean-policy support, and ideal-supported feedback has positive $m_\theta(f \mid D)$. A full-support, uniformly score-bounded softmax actor is one sufficient way to meet the score-support requirements.

(A2) Strongly convex posterior tracking. Auxiliary parameters lie in a compact convex set K. For every $\theta \in \Theta_0$, $\ell(\cdot;\theta)$ is differentiable and uniformly μ-strongly convex on K, with a unique minimizer $\phi^\star(\theta) \in K$ exactly realizing $r_\theta(\cdot \mid D, f)$ for every ideal-supported (D, f). The minimizer map is $L_\star$-Lipschitz.

(A3) Sampling regularity. The auxiliary stochastic gradient $\widehat{h}_t$ is conditionally unbiased for $\nabla_\phi \ell(\phi_t; \theta_t)$, with conditional second moment at most M^2. Furthermore, $\delta(\theta, \phi) \le L_r \|\phi - \phi^\star(\theta)\|$, and the actor uses conditionally fresh ideal-feedback rollouts, with score norm at most B.

(A4) Separated steps. Positive deterministic steps α_t, β_t satisfy

$$\sum_t \alpha_t = \sum_t \beta_t = \infty, \quad \sum_t (\alpha_t^2 + \beta_t^2) < \infty, \quad \frac{\alpha_t}{\beta_t} \to 0, \quad \sum_t \frac{\alpha_t^2}{\beta_t} < \infty. \tag{36}$$

For example $\beta_t \asymp (t+1)^{-0.6}$ and $\alpha_t \asymp (t+1)^{-0.9}$ satisfy these conditions.

Theorem 8.1 (Almost-sure stationarity of projected two-timescale Online GLFT)**.** *Under assumptions (A1)–(A4) above, define $\widehat{h}_t := \nabla_\phi \widehat{\ell}_t^{\,\mathrm{fb}}(\phi_t)$, where $\widehat{\ell}_t^{\,\mathrm{fb}}$ is the per-episode loss in Algorithm 2. The* projected *auxiliary-SGD update and the same clean-policy SGD update,*

$$\phi_{t+1} = \Pi_K \left[\phi_t - \beta_t \widehat{h}_t\right], \qquad \theta_{t+1} = \theta_t + \alpha_t s_{\theta_t}(X_t^+, D_t'),$$

satisfy, almost surely,

$$\|\phi_t - \phi^\star(\theta_t)\| \longrightarrow 0, \tag{37}$$

$$\sum_t \beta_t \|\phi_t - \phi^\star(\theta_t)\|^2 < \infty, \tag{38}$$

$$\sum_t \alpha_t \|\nabla J(\theta_t)\|^2 < \infty, \tag{39}$$

$$\|\nabla J(\theta_t)\| \longrightarrow 0. \tag{40}$$

Thus every limit point of the clean-policy parameters is stationary for J. The theorem does not *establish $p_{\theta_t}(\mathcal{C}(D) \mid D) \to 1$, monotonic task success, or polynomial interaction complexity.*

Proof. The proof is given in Appendix A. Its two steps are posterior tracking by a Robbins–Siegmund drift argument, followed by stochastic ascent with a summable gradient-bias term. The conditions are stated in full because the original population assumptions alone do not imply them. □

8.3 What the result does and does not say

In a finite tabular conditional model with positive trajectory and feedback probabilities, direct probability-vector auxiliary parameters in a compact product of simplexes give a uniformly strongly convex population negative conditional log-likelihood: its Hessian is diagonal with entries proportional to positive joint probabilities divided by squared model probabilities. This verifies that the auxiliary assumptions are *nonvacuous*, but the representation may be exponentially large.

For truthful partial audits from Section 6, many feedback messages have zero likelihood after incompatible executions. The same interior and uniform-strong-convexity argument is not automatic; neither raw-language neural conditional imitation nor efficient posterior estimation is proved by the theorem.

Finally, a restricted actor class may have stationary policies that are not optimal even when feedback is perfectly separating. For a one-step success/failure task with deterministic binary feedback and ideal feedback always equal to "success," set $p_\theta(\text{success}) = \sigma(\theta^2)$. Then $J(\theta) = \log \sigma(\theta^2)$, so $\nabla J(0) = 0$ while success probability is only $1/2$. Hence the original GLFT optimality-separation condition alone cannot upgrade neural-parameter stationarity to task success.

Research boundary. The original exact GLFT operator achieves global feedback matching and, under separation, correct behavior. The projected auxiliary-SGD version of Online GLFT achieves a more conventional *stationary-point* guarantee under additional smoothness, trackability, and timescale conditions; the unprojected version is covered only when the projection would stay inactive. Establishing posterior tracking for realistic free-form language using polynomially many sampled interactions, and eliminating suboptimal neural stationary points, remain separate open questions.

9 Extensions, failure examples, and scope

Distributional separation, not merely pairwise separation. Let one optimal execution produce "A" and "B" with equal probability, while two suboptimal executions produce "A" and "B" deterministically, respectively. Each bad execution has feedback different from the optimal execution, but the equal mixture of the two bad executions matches ideal feedback exactly. The score condition excludes this mixture-level aliasing.

No per-iteration task-success monotonicity, even with $\gamma > 0$. Consider three feasible executions x_A, x_B, x_C, of which x_A, x_B are optimal and x_C is suboptimal. The feedback alphabet is $\{f_A, f_B, f_0\}$, and the feedback channel, with columns in that order, is

	f_A	f_B	f_0	$\mathbb{E}[s(F) \mid x]$
x_A (optimal)	0.8	0	0.2	0.8
x_B (optimal)	0	0.8	0.2	0.8
x_C (suboptimal)	0	0.7	0.3	0.7

Here $s(f_A) = s(f_B) = 1$ and $s(f_0) = 0$. Thus Definition 5.1 holds with $a = 0.8$ and $\gamma = 0.1$. Take $G^\star = F(\cdot \mid x_B)$ and $p_0(x_A, x_B, x_C) = (0.98, 0.001, 0.019)$. Calibration, initial coverage, and exact realizability all hold in a deterministic one-decision environment. Applying the exact update (10) yields

$$p_0(\mathcal{C}) = 0.981, \qquad p_1 \approx (0.19416,\ 0.04559,\ 0.76026),$$
$$p_1(\mathcal{C}) \approx 0.23974, \quad \Phi_0 \approx 3.22886, \quad \Phi_1 \approx 0.20864.$$

Success therefore falls from 98.1% to about 24.0% while feedback KL decreases. Theorem 5.3 nevertheless ensures $p_t(\mathcal{C}) \to 1$ as $t \to \infty$. This example establishes that *monotone feedback*

matching and asymptotic optimality do not require, or imply, monotone policy improvement. The numerical values follow directly from (10), not from a simulation or a different learning algorithm.

Task averaging. The above results are conditional on a fixed request. If $D \sim q^\star$, shared-policy exact population learning induces the request-wise update for $q^\star$-almost every D under the stated conditional realizability assumptions (and at every positive-mass D when $q^\star$ is discrete). For request-dependent $\gamma_D > 0$, average binary error is bounded by $\mathbb{E}_{D\sim q^\star}[\gamma_D^{-1}\sqrt{V_{0,D}/(2(T+1))}]$, provided the expectation is finite. If $\gamma_D \geq \gamma_0 > 0$, Jensen yields $\mathbb{E}_D p_T(\mathcal{C}(D)^c \mid D) \leq \gamma_0^{-1}\sqrt{\mathbb{E}_D V_{0,D}/(2(T+1))}$. No gradient or representation sample-complexity bound is implied by this population averaging.

Approximate ideal-feedback calibration: a qualified extension. Suppose the available generator $\widetilde{G}$ is within ϵ_G total variation of $G^\star$ and is itself realizable as $\widetilde{G} = F\widetilde{p}$ for a distribution $\widetilde{p}$ covered by p_0 (not necessarily optimal). Exact population GLFT using $\widetilde{G}$ obeys Theorem 4.2 with comparator $\widetilde{p}$. Under the original γ-separation of $G^\star$, the triangle inequality gives

$$p_T(\mathcal{C}^c) \leq \frac{1}{\gamma}\left[\epsilon_G + \sqrt{\frac{D_{\mathrm{KL}}(\widetilde{p}\|p_0)}{2(T+1)}}\right]. \tag{41}$$

This is not a guarantee for an arbitrary unachievable generator $\widetilde{G}$: its realizability and coverage are additional assumptions.

Why access to $G^\star$ matters. The ability to sample plausible positive task-relative feedback before observing a successful trajectory is valuable supervision. An LLM or task verifier generating "all requested conditions were met" may be easy to prompt but difficult to calibrate faithfully. The population theorem cannot establish a fair sample-efficiency advantage over a scalar learner unless this access and its construction cost are accounted for.

Support and first-success limitations. If p_0 assigns zero probability to every optimal execution, the multiplicative update cannot create optimal support. In the deterministic full-audit limit, conditioning on wholly positive feedback is equivalent to selecting successful trajectories; a finite dataset may never contain one. Random partial audits permit overlap of feedback between correct and incorrect executions, but the present theorem does not establish an efficient finite-sample learning rate for that case. The exact-population proof is therefore not an end-to-end polynomial-interaction guarantee.

Representations and approximation. Stage-I posterior realizability and Stage-II distillation realizability are strong population conditions. Natural-language teachers may omit crucial requirements, paraphrases may be difficult to ground, and a fixed neural architecture may be unable to represent the posterior. Ordinary approximate-MLE and rollout sampling errors require additional analysis. Our finite-execution theorem does not establish convergence of policy parameters, generalization to unobserved requests, or identification of a particular optimal mixture.

10 Conclusion

General Language Feedback Training has a clean exact-population mechanism: supervised learning fits an execution posterior conditioned on feedback, ideal-feedback sampling changes which executions are emphasized, and distillation produces a request-only policy. Feedback matching is guaranteed under calibration and coverage for the exact population update. Behavioral optimality additionally follows when feedback has a quantitative statistical signal distinguishing all optimal executions from suboptimal mixtures. We derive the gradient of the same feedback-matching objective, show that a fresh first distillation gradient equals it, and prove a separate, conditional stationarity theorem for continuous two-timescale stochastic updates with projected auxiliary SGD. The online theorem

does not inherit the exact operator's global task-optimality guarantee. Truthful random checks of task requirements provide one such channel while allowing substantial omissions, reordering, and paraphrasing. *The convergence argument does not require task success to increase at every iteration, as shown by an example satisfying the paper's own optimality-separation condition.* The remaining practical burden is to learn and distill these conditional distributions from finite data, and to account for the information supplied by the ideal-feedback generator.

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A Detailed proof of Online GLFT convergence

We give the complete stochastic-approximation argument, keeping the two stochastic learning stages and their conditioning separate. Let $\mathcal{F}_t$ be the sigma-field generated by all randomness before the stage-I sample at round t. In particular, θ_t and ϕ_t are $\mathcal{F}_t$-measurable. Define

$$e_t := \phi_t - \phi^\star(\theta_t), \qquad \widetilde{e}_{t+1} := \phi_{t+1} - \phi^\star(\theta_t).$$

Step 1: tracking the moving auxiliary optimum. Projection onto K is nonexpansive; its population minimizer obeys the first-order variational inequality and, by μ-strong convexity,

$$\langle \phi_t - \phi^\star(\theta_t), \nabla_\phi \ell(\phi_t; \theta_t) \rangle \geq \mu \|e_t\|^2.$$

Consequently conditional unbiasedness and the second-moment bound give

$$\mathbb{E}[\|\widetilde{e}_{t+1}\|^2 \mid \mathcal{F}_t] \leq (1 - 2\mu\beta_t)\|e_t\|^2 + M^2\beta_t^2. \tag{42}$$

The actor moves by at most $B\alpha_t$ pathwise and the target minimizer is $L_\star$-Lipschitz, so $\|\phi^\star(\theta_{t+1}) - \phi^\star(\theta_t)\| \leq L_\star B\alpha_t$. Applying $\|u+v\|^2 \leq (1+c\beta_t)\|u\|^2 + (1+1/(c\beta_t))\|v\|^2$ with $c = \mu/2$, for all sufficiently large t we obtain

$$\mathbb{E}[\|e_{t+1}\|^2 \mid \mathcal{F}_t] \leq (1 - \mu\beta_t)\|e_t\|^2 + C_1\beta_t^2 + C_2\frac{\alpha_t^2}{\beta_t}. \tag{43}$$

This inequality does not require independent stage-I and stage-II *targets*; the pathwise actor-step bound suffices. The Robbins–Siegmund almost-supermartingale lemma [5] and (A4) imply that $\|e_t\|^2$ converges and $\sum_t \beta_t \|e_t\|^2 < \infty$ almost surely. Since $\sum_t \beta_t = \infty$, its limit must be zero. This proves (37) and (38).

Step 2: bounding the actor-gradient bias. After stage I, define the actor-score conditional mean and deviation

$$g(\theta_t, \phi_{t+1}) = \mathbb{E}[s_{\theta_t}(X_t^+, D_t') \mid \mathcal{F}_t, \phi_{t+1}], \quad b_t = g(\theta_t, \phi_{t+1}) - \nabla J(\theta_t).$$

Proposition 7.2, (A3), and (42) give, for a constant C,

$$\mathbb{E}[\|b_t\|^2 \mid \mathcal{F}_t] \le 4B^2 L_r^2 \mathbb{E}[\|\widetilde{e}_{t+1}\|^2 \mid \mathcal{F}_t] \le C(\|e_t\|^2 + \beta_t^2). \tag{44}$$

As $\alpha_t \le \beta_t$ eventually, the tracking sum and square-summable steps imply

$$\sum_t \alpha_t \mathbb{E}[\|b_t\|^2 \mid \mathcal{F}_t] < \infty \qquad \text{almost surely.} \tag{45}$$

Step 3: stochastic ascent and stationary gradients. Use the smooth ascent inequality on each actor update segment, condition first on $\mathcal{F}_t$ and then on the auxiliary stage-I draw, and use $\|s_{\theta_t}\| \le B$:

$$\mathbb{E}[J(\theta_{t+1}) \mid \mathcal{F}_t] \ge J(\theta_t) + \alpha_t \|\nabla J(\theta_t)\|^2 + \alpha_t \mathbb{E}[\langle \nabla J(\theta_t), b_t \rangle \mid \mathcal{F}_t] - \tfrac{L_J B^2 \alpha_t^2}{2} \tag{46}$$

$$\ge J(\theta_t) + \tfrac{\alpha_t}{2} \|\nabla J(\theta_t)\|^2 - \tfrac{\alpha_t}{2} \mathbb{E}[\|b_t\|^2 \mid \mathcal{F}_t] - \tfrac{L_J B^2 \alpha_t^2}{2}, \tag{47}$$

where the second line uses the elementary Young inequality $\langle u, v \rangle \ge -\|u\|^2/2 - \|v\|^2/2$. Because $J \le 0$, the nonnegative process $-J(\theta_t)$ satisfies a Robbins–Siegmund inequality with summable error terms by (45) and (A4). Thus $\sum_t \alpha_t \|\nabla J(\theta_t)\|^2 < \infty$ almost surely, proving (39).

For completeness, this weighted-summability conclusion plus bounded increments implies actual norm convergence rather than merely a vanishing liminf. Set $u_t = \|\nabla J(\theta_t)\|$. The L_J-Lipschitz gradient and bounded actor score give $|u_{t+1} - u_t| \le L_J B \alpha_t$. If $u_t \ge \varepsilon > 0$ at arbitrarily late times, either $u_t \ge \varepsilon/2$ from some time onward (contradicting $\sum \alpha_t u_t^2 < \infty$ and $\sum \alpha_t = \infty$), or infinitely many disjoint excursions from below $\varepsilon/2$ to at least ε occur. Every such excursion has $\sum \alpha_t \ge \varepsilon/(2L_J B)$ and contributes at least $\varepsilon^3/(8L_J B)$ to $\sum \alpha_t u_t^2$, again a contradiction. Hence $u_t \to 0$ almost surely. Continuity of ∇J and boundedness of the iterates imply every accumulation point is stationary. □

B Additional audit of the gradient formulation

Exact conditional sampling is statistically well defined, but can be costly. Given an ideal feedback sample $f^\star$, repeatedly draw $X \sim p_\theta(\cdot \mid D)$ and $F \sim F_D(\cdot \mid X)$ until $F = f^\star$. The accepted X has law $r_\theta(\cdot \mid D, f^\star)$ because one-trial acceptance probability for $X = x$ equals $p_\theta(x \mid D) F_D(f^\star \mid x)$, whose sum is $m_\theta(f^\star \mid D)$. The conditional expected trials are $1/m_\theta(f^\star \mid D)$, so the average cost is

$$\mathbb{E}_{D \sim q^\star,\, f^\star \sim G_D^\star} \frac{1}{m_\theta(f^\star \mid D)}.$$

This gives an exact unbiased gradient implementation *using only the sampling resources postulated by GLFT*, but offers no polynomial rollout guarantee and is especially problematic with raw, high-entropy feedback strings.

A finite-time approximate-gradient statement. For an actor-only implementation with fresh conditional samples from an auxiliary model of average TV error δ_t, assume J is L-smooth along actor steps, the trajectory score is bounded by B, and $\theta_{t+1} = \theta_t + \alpha s_{\theta_t}(X_t^+, D_t)$ for constant $\alpha > 0$. Repeating the conditional-bias bound and smooth-ascent argument above, without needing the auxiliary tracking theorem, yields

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\|\nabla J(\theta_t)\|^2 \le \frac{2[-J(\theta_0)]}{\alpha T} + \frac{4B^2}{T}\sum_{t=0}^{T-1} \mathbb{E}\delta_t^2 + L\alpha B^2. \tag{48}$$

This is a *conditional finite-iteration stationarity bound*, not a bound on how many feedback examples are needed to achieve a given δ_t. Exact conditional sampling and $\alpha \asymp T^{-1/2}$ give an $O(T^{-1/2})$ average squared-gradient bound, not global optimality.

Why this does not yet solve the original sample-complexity question. Our new online theorem assumes $\ell(\cdot;\theta)$ is uniformly strongly convex and posterior-realizable in a parameter class with controlled score norms and bounded iterates. For tabular full-support conditional probability vectors these can hold when the parameter domain is appropriately bounded away from its boundary and contains the required posteriors; the constants and representation size may nevertheless depend exponentially on horizon. Natural-language partial auditors, unrestricted neural networks, off-policy replay, and jointly learned ideal-feedback generators are not covered. Without additional assumptions ruling out suboptimal stationary points, GLFT's γ-separation translates convergence of *feedback distributions* to correctness, not convergence of neural gradients to correctness.

C Audit notes and proof-boundary checklist

1. **Bayes step.** The Stage-I conditional likelihood identifies the full posterior $p_t(x)F(f \mid x)/m_t(f)$ only under policy realizability. With uncontrolled stochastic transitions, the posterior may violate the environment's transition law.

2. **Support.** Calibration and $\mathrm{supp}(\bar{p}) \subseteq \mathrm{supp}(p_0)$ imply all $G^\star$-positive messages have positive current mass, and the multiplicative update preserves $\bar{p}$ support. This justifies denominators and finite comparator KL at each finite t.

3. **Monotonicity.** The two inequalities in (12) use Jensen in different directions of conditioning, plus exact posterior sampling. Neither follows for arbitrary approximate neural training. As the explicit example in Section 9 shows, they do not imply monotone task-success probability, even when γ-optimality separation holds.

4. **Score gap.** The score must have the same mean a on all optimal executions; an inequality only comparing individual pairs of trajectory feedback is insufficient for mixtures.

5. **Truthful auditor.** Independence *between* audited requirements is not needed. The entire subset-selection law Q_D, not merely each requirement's marginal inclusion probability, must be execution-independent to make request-only ideal feedback calibrated for every optimal execution. Inclusion probabilities must be strictly positive, all reported polarities semantically correct, and the wording kernel the same for the actual and ideal feedback processes. The analysis is invariant to permutations of the report chunks.

6. **Rates.** $O(1/T)$ is a KL feedback-distribution rate, while binary success error is bounded by $O(T^{-1/2})$. Neither is a finite-rollout rate; the constants include initial inverse coverage through $D_{\mathrm{KL}}(\bar{p}\|p_0)$.

7. **Ideal-feedback access.** Generation is an explicit training-time resource and must not be silently granted to one side of a sample-efficiency comparison.

8. **Online SGD.** Algorithm 2 displays ordinary SGD on both imitation losses. Theorem 8.1 establishes stationary-point convergence for the projected auxiliary-SGD variant under strong additional auxiliary optimization and smoothness assumptions; it covers unprojected updates only when projection is inactive, not generic neural GLFT training.

9. **Reward objective.** Exact optimality here means satisfying all required constraints; the graded-quality corollary handles the mean fraction satisfied. Near-optimal reward objectives require the separate ε-margin in (19).